IACE PLATFORM · PRODUCT

What the platform does today

The product as it is actually built, feature by feature, with what is finished, what is dark, and what was deliberately left out. Written from the code, not from the plan.

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Surface 225 endpoints, 3 clients, 53 reports
Companion docs/engineering-handbook.pdf, for how each of these works

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What this is

In one paragraph

A mock-test platform for a government-exam coaching institute. Staff build papers from a reusable blueprint, either by writing the questions or by drawing them from a bank, and two people type and proof-read every paper before it is offered. Students sit those papers under exam conditions, on a clock, in a screen made to look like the real exam's, and are marked within seconds of handing in. They then get a score card, their rank and percentile against everyone who sat the same paper, a worked solution for every question, and a record of how they are trending. Staff get a live board for test day and 53 reports. It runs on the web for staff, on the web for students, and on an Android app that is built but not launched.

The three clients

Client	Who uses it	State
Admin portal — `apps/admin`	Institute staff: owners, typists, proof-readers	Built. 176 of the 225 endpoints are its
Student portal — `apps/exams`	Candidates, on a laptop or phone browser	Built, and the only way to sit a paper today
Mobile app — `apps/mobile`	Candidates, Android first	Screens complete and pointed at staging; not launched, push unproven on a device

What the product has to survive

One number shapes almost every technical decision: three thousand students sitting the same paper in the same minute, with headroom to eight thousand. A mock test is a synchronised event — everybody opens at the bell and most hand in within a few minutes of each other. That is why the timer runs in the browser, why answers are held in memory and written to the database on a timer rather than per keystroke, and why marking happens on a queue after the fact rather than in the request that submits the paper.

WHAT A STUDENT DOES, END TO END

Before

Sign in with a code → by SMS · See what is open → catalog · Read the rules → the bell

During

Sit the paper → autosaved every ~25 s · Hand in → marked on a queue

After

Score card → rank, percentile · Solutions → worked, per question

Leaderboard → podium, then every seat · Downloads → own reports as PDF or sheet

The student's product

Getting in

A student has no password and no PIN. They sign in with their mobile number and a one-time code sent to it by SMS, every time, and the first time a number does it is the signup. A code can be re-sent after a countdown. A student whose SMS never arrives can be read a single-use desk code by an admin, valid five minutes and never sent anywhere.

> **Note** An old mobile number is never a way in. Operators recycle numbers, so a number belongs to whoever holds it now. Only an admin can change a student's number, and doing so signs the student out everywhere.

A student gets one session per kind of app: signing in on a second phone ends the first phone's session, while a browser session continues alongside it. A tab that was signed out by another device says which kind of device did it.

> **Note** WhatsApp delivery is built and switched off. Codes go by SMS alone.

What a student can see

A student does not browse a catalogue of everything the institute sells. They see exactly the test series that reach them, by one of five routes:

How a series reaches a student	What it depends on
Free	Reaches every active student
Standard	Their branch is on the series' list and their course matches
Program	They hold the program
Event	They are on that event's candidate roster
Grant	An admin gave this one student this one series

A test's own status is the one switch: active or inactive. A series can be sequential, so its tests must be sat in order; a retake never re-locks the ones after it. Individual tests have an opening time, and a program can be given an earlier one. A test that cannot be started says why: held, not yet open, or waiting its turn.

> **Note** This is recomputed on every read rather than stored, so a change to a student's branch, programs or grants takes effect on their next screen refresh.

Sitting a paper

The sitting is the product. It replicates a government computer-based test closely enough that the hall is no surprise: a question panel, an option list, a palette of every seat colour-coded by state, a section bar and a countdown. Four skins exist: the default, Railways, Banking and SSC, the last three copied from the real exam screens.

What it does:

- **The paper is fetched before the student presses Begin**, so the bell does not become a stampede on the server.
- **The clock is granted by the server and counted down by the browser.** The device clock is never trusted. At zero the paper locks, and an ended sitting says "This test has ended".
- **Answers are kept locally and sent in batches** roughly every 25 seconds. A failed send never takes an answer off the screen, and an offline tab keeps its queue and resends on reconnect.
- **One sitting, one sign-in.** A second device trying to answer the same sitting gets a hold screen and Try again until the first has been quiet for a minute. A reload resumes the same sitting and never starts a new one.
- **Pause, and section hand-in.** A student can pause, and on sectional papers hand in a section before its clock runs out.
- **Forward-only papers** are enforced by the screen and the server: a student cannot change an answer behind the furthest seat reached, and has no mark-for-review.
- **Question kinds:** single-choice, a typed answer, a typing passage, and a descriptive answer marked by rule against key points. A typing practice area sits outside tests.

> **Watch** Two honest limits. If a student answers on two devices at once, the second device's queue can overwrite the first's; the fix is a per-question version on the server. And a sectional paper's section clocks are enforced by the screen only.

Getting marked

Handing in is one request. The server records the submission and queues the marking; the student lands on a handed-in page that compares their effort with the cohort before the marks exist.

Marking happens on a worker: it reads the answer sheet, compares each answer to the key, applies the paper's marks and negative marks, and writes the result. A question the staff later drop gives its marks back to anyone who attempted it; a bonus question gives marks to everyone. Changing a question's disposition re-marks every sitting of that paper within about two minutes, and a student whose total moves is told "Your result has been updated". If marking fails or is lost, a sweep notices within two minutes and asks again.

Reading the result

Screen	What it answers
Score card	What did I score, out of what, and where does that put me
Sections	Which parts of the paper carried me and which cost me
Solutions	Why each answer is right — stem, options and a worked solution
Questions	Every question with my answer, the correct one, my time, and how the cohort did
Compare	Me against the cohort and against the topper
Leaderboard	The podium of three, then every seat twenty to a page, with mine pinned
Performance	My trend across sittings, and my accuracy by subject

A typing paper shows a typing result instead of marks. A sitting an admin set aside says so on every tab.

Rank and percentile are counted live. They are not stored and there is no regeneration step: the moment the next student is marked, every rank is correct.

> **Watch** A student whose paper is marked can open the answer key while others are still writing, and can pass it on. The gate is "your own sitting, and it has been marked". This is a deliberate trade, and worth re-deciding before a public rollout rather than after.

The rest of the student's account

Saved questions, an in-app notification bell, a profile with a photo and a 10th marksheet, a list of signed-in devices with the ability to sign one out, and a Downloads tab with five reports about themselves as PDF or spreadsheet. Announcements and "test opened" reach the bell the moment they are sent, and a student never sees one sent before they joined. Identity documents are never stored: Aadhaar and PAN exist only as verification flags, and nothing can set them today.

The institute's product

Who the staff are

Staff are not students with extra rights; they are a separate kind of account entirely, and they sign in with an emailed one-time code every time. What a staff member may do is a set of named permissions, each at read, write or export level. A role — typist, proof-reader, admin, super admin — seeds those permissions, and changing a role prunes the ones the new role lacks, warning with figures before it does.

A screen a staff member has no permission for is refused when they open it, rather than failing on save; a screen held at read opens read-only. The last active super admin cannot be demoted or deactivated, and nobody can deactivate themselves. Branches, exams, stages and programs are written by super admins alone, and so is the audit log.

Building a paper

This is the deepest part of the product and the part most unlike an off-the-shelf tool.

A blueprint, not a template. A base config describes the shape of an exam paper: its sections, how many questions each holds, the marks and negative marks, the duration, how the timer behaves (whole paper, per section, or per question), which languages, and which on-screen skin. A section can be a typing test or a written one. Tests are cut from a blueprint, and many tests share one.

Two ways to fill a paper:

	Picked	Framed
Where questions come from	Drawn from the bank	Written for this paper
Who fills it	The test owner, by drawing or hand-picking	An assigned typist
The draw	Seeded fresh each time, honouring a topic and difficulty mix	—

Every paper is read before it is offered. A section is assigned to a typist and a proof-reader with a due day. The typist writes or chooses, with a clock on each question; pressing Done hands the section to the reader. The reader ticks each question, or sends one back with a reason; a sent-back question becomes editable again, even after Done. A whole typed test can be imported from one spreadsheet.

The offer has gates. A test cannot be offered until every section's work is finished, every scoped section has a proof-reader, and every question carries its tick, unless a super admin overrides the reading gates. Missing languages block the offer unless an admin waives them by name; a waived language is not offered to candidates.

Offering freezes the paper. From that moment the database itself refuses changes: triggers, not application code, stop a paper or a question's content moving once anybody has sat it. The one thing that can still change is a question's disposition — dropping or bonusing it — which re-marks every sitting.

> **Watch** A frozen paper cannot be unfrozen. A test can be made inactive, which hides it and stops new sittings, but the paper stays frozen permanently.

The question bank

Questions live under a subject and topic taxonomy with free-text tags, in up to three languages (English, Hindi, Telugu), with figures and mathematical formulas. A language subject, such as Hindi comprehension, is written in its own language alone. The list sorts and filters by tag, author, date and language, and exports as a background job.

Each question has versions: a paper pins the exact version it served, so correcting a typo can never change what a student already sat. An edit rewrites the existing version only when no paper that pins it is reachable by a student; otherwise a new version is written. That is the single most consequential rule in the bank, because answer sheets store option positions rather than option ids.

Questions can be imported in bulk from a spreadsheet. The preview counts each kind of problem once, lets the admin map a missing subject, topic or difficulty to a bank value or create it, and lets single rows be corrected or left out. Only valid rows are written.

Running a test day

Capability	What it does
Live board	Watch a test in progress — who has started, who is still writing
Force submit	End one student's sitting
Extend	Give one student more time; it can never shorten a clock
Reset / void	Let a student sit again, or set a sitting aside from the rankings
Analytics	Per-test and per-question performance across the cohort
Announcements	A message to a filtered set of students, in-app and optionally by SMS, with a cost estimate before sending
Audit	Who changed what, with before-and-after values, for super admins; rows older than 30 days are archived

Reports

Fifty-three reports live in one catalogue, searchable and grouped, each on its own page with its filters in the address. Every report prints with the institute's logo and exports as a spreadsheet built in the background; the parent letter and the printed question paper are reports too. Four of them — the answer key, the audit trail, admin activity and the permissions matrix — are a super admin's alone. Printing stops at 5,000 rows a table; the spreadsheet carries every row.

> **Watch** Reports are institute-wide: whoever holds the reports permission reads every branch. If a branch office should see only its own branch, that is a product decision still to make. Reading a report is not audited; exporting one is.

Roster and catalog

Students are added one at a time or imported by spreadsheet, with their branch, courses and programs. A student who signs up on their own arrives as a non-IACE student. Branches are physical or virtual. A student can be deactivated (cannot sign in, and is signed out at once), blocked from tests (can sign in, cannot start one, and cannot be granted a series), or erased, which anonymises their

personal data in place while keeping their sittings, because a cohort's statistics must not move when one person exercises their rights. Erasure removes the current photo and marksheet from storage before the row.

> **Watch** Erasure does not yet reach two places: earlier photos and marksheets a student replaced, and the original roster spreadsheet the student was imported from. Both stay in storage.

Events carry their own candidate roster, imported by spreadsheet. Retiring an event hides it from pickers but does not stop existing series reaching its candidates.

What is not built

Honest scope. Each of these is a deliberate absence, not an oversight.

Not built	Status
The mobile app's launch	Screens complete, the sitting included; a preview build runs against staging. Typing and written papers are refused on a phone. The sitting has never been verified on a physical device, and the production build has no server address yet
Mobile push notifications	Both ends are built; no notification has been seen to reach a device. The in-app bell works everywhere
WhatsApp delivery	Built and switched off
Production	Staging runs on AWS and is switched off between test sessions. No production environment exists
A broad student portal	V1 is the mock-test feature. Course content, doubt-solving and the rest are later
Identity verification	Aadhaar and PAN flags exist; nothing can set them. Images are deliberately never stored
A solutions release gate	Removed on purpose; see the note above
Re-scoring one sitting on demand	The only lever is a question's disposition, which re-marks the whole cohort

What stands between this and a launch

- **The load test.** The 6,000-student test has not been run against staging.
- **The mobile app on a real phone.** Nothing past sign-in and uploads has been tried on a device.
- **Three product calls.** Whether a branch office should see only its own branch in reports; whether a retired event should stop reaching its candidates; and whether the answer key should wait for the whole batch to finish.